Frequency response

Answers and worked solutions · Day 07 · Fri 30 Oct · pack D4

5. Practice

From your course files

T4 Q3 (tutorial).

Worked as Example 4.

(a) $H(\omega) = \dfrac{Y(\omega)}{X(\omega)} = \dfrac{\mathbf{1 - \frac{1}{2}e^{-j\omega}}}{\mathbf{1 - \frac{1}{8}e^{-j\omega}}}$

(b) $e^{j\omega n}$ is an eigenfunction of LTI systems, so $\mathbf{H(\pi/5) = 3e^{-j\pi/3}}$ only. $H(\omega)$ at other frequencies cannot be found from this pair.

(c) Spot that the input is $\delta(n)$, so $h(n) = \left(\frac{1}{5}\right)^{n+2}u(n+2)$ and $\mathbf{H(\omega)} = \dfrac{\mathbf{e^{j2\omega}}}{\mathbf{1 - \frac{1}{5}e^{-j\omega}}}$.

T4 Q5 (tutorial).

Input frequency in every case: $\pi/4$ rad. Hint: reduce $5\pi/4$ to $[0, \pi]$.

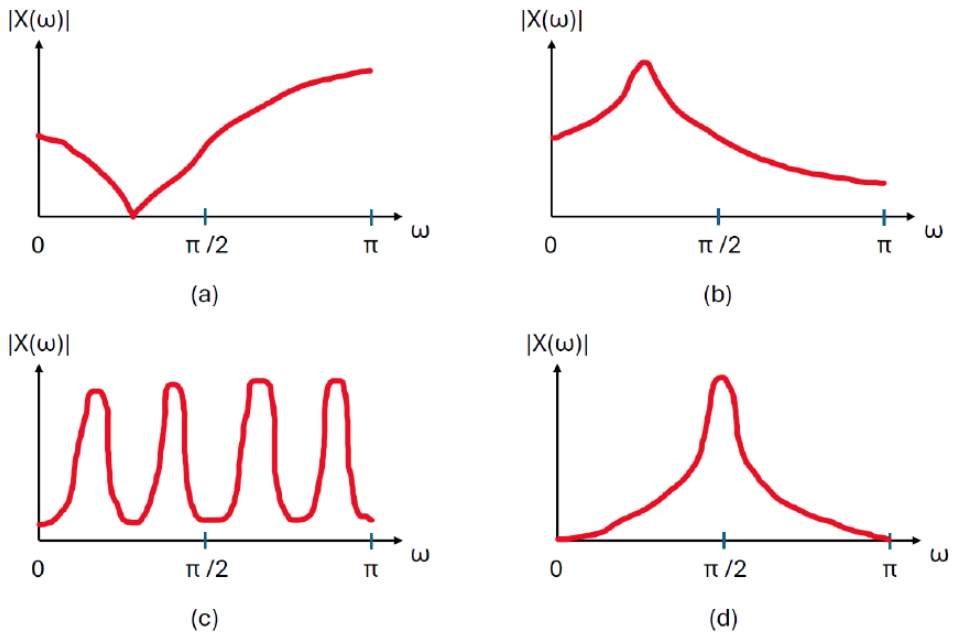
(a) $y(n) = A\cos(\frac{\pi}{2}n)$. Output frequency $\pi/2$ rad, so the system is **time-varying**.

(b) $y(n) = A^2\cos^2(\frac{\pi}{4}n) = \frac{1}{2}A^2 + \frac{1}{2}A^2\cos(\frac{\pi}{2}n)$. Output frequencies 0 and $\pi/2$ rad, so the system is **nonlinear**.

(c) $y(n) = A\cos(\frac{\pi}{4}n)\cos(\pi n) = A\cos(\frac{3\pi}{4}n)$. Output frequency $3\pi/4$ rad, so the system is **time-varying**.

None of the three is LTI.

T4 Q6 (tutorial).



T4 Q6: the official sketches (from the tutorial solutions)

Pole and zero positions below are estimated from the plots. Numbers in brackets come from evaluating those estimated patterns in Python. They are a guide to the shape, not official values.

The course solution says: walk round the unit circle. $|X|$ goes small near a zero and large near a pole. Poles or zeros at the origin have no effect on magnitude. Sketch $0 \le \omega \le \pi$.

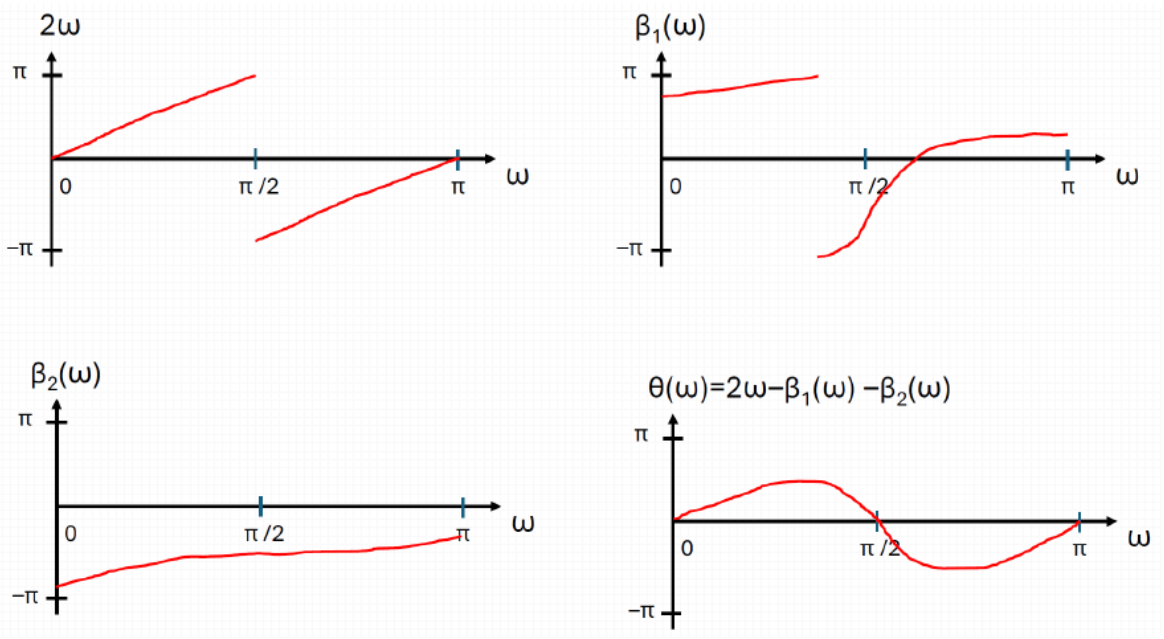
(a) Double pole at the origin. Zeros on (or very near) the unit circle at $e^{\pm j\pi/4}$. This is a second-order FIR notch, like $1 - \sqrt{2}e^{-j\omega} + e^{-j2\omega}$. Shape: it starts at a moderate value at $\omega = 0$ (about 0.59) and falls to **zero at** $\omega \approx \pi/4$. It then rises steadily and smoothly to its **maximum at** $\omega = \pi$ (about 3.41), where $e^{j\omega}$ is furthest from both zeros. It looks like a "V" notch near $\pi/4$ followed by a long rising curve.

(b) Double zero at the origin. Complex-conjugate poles inside the circle at about $0.9e^{\pm j\pi/4}$. Shape: a moderate value at $\omega = 0$, rising to a **single rounded peak at** $\omega \approx \pi/4$, then decaying smoothly to a small value at $\omega = \pi$. A resonance or band-pass shape. It does not reach zero anywhere, because there are no zeros on the circle.

(c) Eight zeros equally spaced on the unit circle at angles $k\pi/4$ ($k = 0, \ldots, 7$, including $z = 1$ and $z = -1$). An 8th-order pole at the origin. This is $1 - e^{-j8\omega}$, so $|X| = 2|\sin 4\omega|$. Shape: **zero at** $\omega = 0, \pi/4, \pi/2, 3\pi/4, \pi$, with **four equal humps** in between. The humps peak at $\pi/8, 3\pi/8, 5\pi/8, 7\pi/8$, each with height 2. It is a comb filter.

(d) Zeros on the circle at $z = \pm 1$. Poles at about $\pm j0.9$. This is like $(1 - e^{-j2\omega})/(1 + 0.81e^{-j2\omega})$. Shape: **zero at** $\omega = 0$, rising slowly and then steeply to a **tall sharp peak at** $\omega = \pi/2$ (about 10.5 for poles at radius 0.9), then falling back to **zero at** $\omega = \pi$. It is symmetric about $\pi/2$. A narrow band-pass shape, the same kind of curve as Figure 1 but centred at $\pi/2$.

T4 Q7 (tutorial).



T4 Q7: the official sketches of 2ω, $\beta_1(\omega)$, $\beta_2(\omega)$ and $\theta(\omega)$ (from the tutorial solutions)

Pattern: a double zero at the origin and two poles at about $\pm j0.9$, with $b_0 = 1$. So $X(\omega) = \dfrac{e^{j2\omega}}{(e^{j\omega} - p_1)(e^{j\omega} - p_2)}$, which is $1/(1 + 0.81e^{-j2\omega})$.

Course method: $\theta(\omega) = 2\omega - \beta_1(\omega) - \beta_2(\omega)$, with $\beta_i = \angle(e^{j\omega} - p_i)$. The solution sketches four small plots: 2ω, β_1, β_2 and the sum. The three pieces are hand-drawn roughly and in wrapped form (angles folded into $(-\pi, \pi]$), so your sub-sketches may look different. What matters is the final $\theta(\omega)$. Here is what the pieces actually do, computed for poles at $\pm j0.9$:

- 2ω: a straight line from 0 up to π at $\omega = \pi/2$. In wrapped form it then jumps to $-\pi$ and rises to 0 at $\omega = \pi$.
- Angle from the pole at $+j0.9$: -42° at $\omega = 0$, rising slowly (-15° at $\pi/4$). It then swings very fast through $+90^\circ$ at $\omega = \pi/2$ (about $29^\circ \to 151^\circ$ between 0.45π and 0.55π), because $e^{j\omega}$ passes right next to this pole. It reaches 180°, wraps to -180°, and ends at -138° at $\omega = \pi$.
- Angle from the pole at $-j0.9$: a slow, smooth rise, $42^\circ \to 90^\circ$ (at $\pi/2$) $\to 138^\circ$ (at π).

Resulting $\theta(\omega)$ (what you should draw): $\theta(0) = 0$. It rises to a positive maximum of about $+0.94$ rad ($\approx 54^\circ$) near $\omega \approx 0.4\pi$. It then falls steeply through **0 at** $\omega = \pi/2$ (where the pole is closest). It reaches a minimum of about -0.94 rad near 0.6π and returns to **0 at** $\omega = \pi$. The curve is odd-symmetric about $\omega = \pi/2$. It looks like one smooth "up-then-down" wiggle, with a steep negative slope at the resonance.

Check: $X(0) = 1/1.81$ and $X(\pi) = 1/1.81$ are both real and positive, so the phase is 0 at both ends. $X(\pi/2) = 1/0.19$ is real and positive, so the phase is 0 there too.

New questions

N1.

a. $H(\omega) = 1 + 2e^{-j\omega} + e^{-j2\omega} = e^{-j\omega}(e^{j\omega} + 2 + e^{-j\omega}) = \mathbf{e^{-j\omega}(2 + 2\cos\omega)}$.
$R(\omega) = 2 + 2\cos\omega \geq 0$, so $|H| = 2 + 2\cos\omega$ and $\theta = -\omega$. This is linear phase.

b. At $\omega = \pi/3$: $|H| = 2 + 1 = 3$ and $\theta = -\pi/3$.

$$y(n) = 12\cos\left(\tfrac{\pi}{3}n + \tfrac{\pi}{4} - \tfrac{\pi}{3}\right) = \mathbf{12\cos\left(\tfrac{\pi}{3}n - \tfrac{\pi}{12}\right)}$$

This was checked by direct convolution.

c. $2 + 2\cos\omega = 0$ gives $\boldsymbol{\omega = \pi}$. That is a double zero at $z = -1$, so $(-1)^n$ is blocked.

N2.

a. $H(\omega) = b_0 + b_1 e^{-j\omega}$. DC ($\omega = 0$): $H(0) = b_0 + b_1 = 3$. $(-1)^n$ ($\omega = \pi$): $H(\pi) = b_0 - b_1 = -1$. Solving gives $\mathbf{b_0 = 1,\ b_1 = 2}$. Without the known filter form, two pairs would give only $H(0)$ and $H(\pi)$ (compare T4 Q3b).

b. $H(\pi/2) = 1 + 2(-j) = 1 - 2j$. So $|H| = \sqrt{5} = 2.236$ and $\theta = -\arctan 2 = -1.107$ rad.

$$y(n) = \mathbf{2.236\cos\left(\tfrac{\pi}{2}n - 1.107\right)}$$

This was checked numerically against $\cos(\frac{\pi}{2}n) + 2\cos(\frac{\pi}{2}(n-1))$.

N3.

a. $y = A^2\cos(\frac{\pi}{3}n)\cos(\frac{\pi}{3}n - \frac{\pi}{3}) = \frac{A^2}{2}\left[\cos\frac{\pi}{3} + \cos(\frac{2\pi}{3}n - \frac{\pi}{3})\right]$. Frequencies **0 and** $2\pi/3$ are new, so the system is **nonlinear** and not LTI.

b. $H(\omega) = 1 + e^{-j2\omega} = 2e^{-j\omega}\cos\omega$. At $\pi/3$: $|H| = 1$ and $\theta = -\pi/3$.
$y = A\cos(\frac{\pi}{3}n - \frac{\pi}{3})$. The only frequency is $\pi/3$. This is consistent with LTI, and the system is in fact LTI (an FIR LCCDE).

c. $y = A\cos(-\frac{\pi}{3}n) = A\cos(\frac{\pi}{3}n)$. **No new frequency**, but the system is **not** LTI. Shifting the input by k gives $x(-n-k)$, while shifting the output gives $x(-(n-k)) = x(-n+k)$, so it is time-varying. The frequency test can't prove LTI.

d. $y = 2A\cos(\frac{\pi}{3}n) + 1$. A **new DC component** ($\omega = 0$) appears, so the system is not LTI. It is not linear, because a zero input gives a non-zero output.

N4.

a. $Y(1+0.81e^{-j2\omega}) = X(1+e^{-j2\omega})$, so

$$H(\omega) = \frac{1+e^{-j2\omega}}{1+0.81e^{-j2\omega}} = \frac{e^{j2\omega}+1}{e^{j2\omega}+0.81}$$

b. Zeros: $z^2 = -1$, so $z = \pm j$ (on the unit circle at $\omega = \pm\pi/2$). Poles: $z^2 = -0.81$, so $p = \pm j0.9$ (inside the circle, so the filter is stable). $b_0 = 1$ and $N = M = 2$, so there is no linear phase term.

c. $e^{j\pi/4} = 0.7071 + j0.7071$.

From	Length	Angle
zero $+j$	$V_1 = 0.7654$	-22.5°
zero $-j$	$V_2 = 1.8478$	67.5°
pole $+j0.9$	$U_1 = 0.7329$	-15.26°
pole $-j0.9$	$U_2 = 1.7558$	66.25°

$$|H(\pi/4)| = \frac{0.7654 \times 1.8478}{0.7329 \times 1.7558} = \mathbf{1.0989}$$

$$\theta(\pi/4) = (-22.5 + 67.5) - (-15.26 + 66.25) = \mathbf{-5.99}^\circ = -0.1046 \text{ rad}$$

d.

- $H(0) = 2/1.81 = 1.1050$, so the DC output is 5.525.
- $H(\pi/2) = 0$ (zero on the circle), so the $\cos(\frac{\pi}{2}n)$ term is **removed**.
- The $\pi/4$ term has amplitude $3 \times 1.0989 = 3.297$ and phase $\pi/3 - 0.1046 = 0.9426$ rad ($\approx 0.300\pi$).

$$y_{ss}(n) = \mathbf{5.525 + 3.297\cos\left(\frac{\pi}{4}n + 0.943\right)}$$

This was checked by simulating the LCCDE.

e. $|H|$ is flat at about 1.1 over most of the band (1.105 at 0 and π, 1.099 at $\pi/4$ and $3\pi/4$). It has a **narrow notch dropping to 0 at** $\omega = \pi/2$. The 3 dB points are at about 0.472π and 0.528π. The poles just behind the zeros are what keep the notch narrow.

N5.

a. $H(0) = b_0 \cdot 2/0.2 = 10b_0 = 1$, so $\mathbf{b_0 = 0.1}$.

b. $H = 0.1\left[\frac{1}{1-0.8e^{-j\omega}} + \frac{e^{-j\omega}}{1-0.8e^{-j\omega}}\right]$, so

$$h(n) = \mathbf{0.1(0.8)^n u(n) + 0.1(0.8)^{n-1}u(n-1)}$$

$\sum h = 0.1\left(\frac{1}{0.2} + \frac{1}{0.2}\right) = 1 = H(0)$, as required.

c. Write $H = 0.1\dfrac{e^{j\omega}+1}{e^{j\omega}-0.8}$, with zero $z_1 = -1$ and pole $p_1 = 0.8$, $N = M = 1$. At $e^{j\pi/2} = j$:

- zero vector: $j - (-1) = 1 + j$, length $\sqrt{2} = 1.4142$, angle 45°;
- pole vector: $j - 0.8 = -0.8 + j$, length 1.2806, angle 128.66°.

$$|H(\pi/2)| = 0.1 \times \frac{1.4142}{1.2806} = \mathbf{0.1104}, \qquad \theta(\pi/2) = 45^\circ - 128.66^\circ = \mathbf{-83.66}^\circ = -1.460 \text{ rad}$$

d. $H(\pi) = 0$ (the zero at $z = -1$), so $(-1)^n$ is removed.

$$y_{ss}(n) = \mathbf{0.1104}\cos\left(\tfrac{\pi}{\mathbf{2}}\mathbf{n} - \mathbf{1.460}\right)$$

This was checked by simulation.

e. $|H|^2 = 0.01\dfrac{2(1+\cos\omega)}{1.64 - 1.6\cos\omega} = \dfrac{1}{2}$. This gives $0.02 + 0.02\cos\omega = 0.82 - 0.8\cos\omega$, so $\cos\omega_c = 0.8/0.82 = 0.9756$ and $\boldsymbol{\omega_c = 0.2213}$ rad $(\approx 0.0704\pi)$. It is a narrow low-pass filter: the pole near $z = 1$ boosts DC and the zero at $z = -1$ kills $\omega = \pi$.

6. Quick recall check

Question

1. $7H(0)$, where $H(0) = \sum_n h(n)$ is real.
2. $2\cos(\omega_0 n + 0.3 + \pi)$, or equivalently $-2\cos(\omega_0 n + 0.3)$. The magnitude is 2 and the phase is π.
3. Zero. That frequency is completely blocked (a notch).
4. $|H|$: no, because its distance to every point on the circle is 1. θ: yes, it adds the linear term $+\omega$ via $\omega(N - M)$.
5. No. An LTI system can't create a new frequency, since $Y = HX$. It must be nonlinear and/or time-varying.